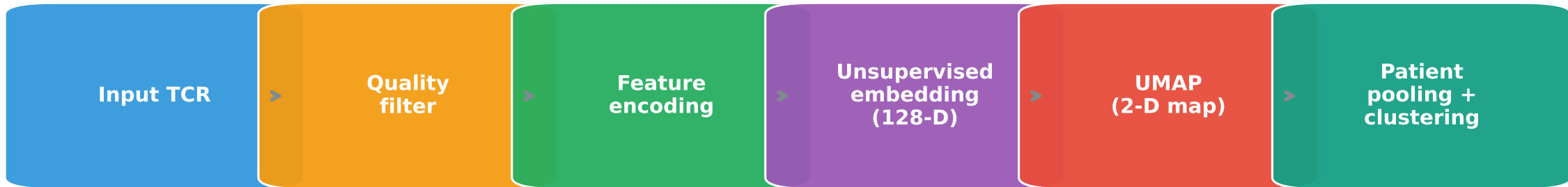


Figure S14: Unsupervised learning pipeline (CDR3 + V/J → embedding → UMAP)



Unsupervised embedding is learned without outcome labels; labels are used only for coloring/interpretation in plots.